QUBO Formulation: Cardinality-Constrained Portfolio Selection Problem

Cardinality-Constrained Portfolio Selection Problem

Problem Statement. Given a set of N assets indexed by $i \in \{1, \ldots, N\}$, each asset i is associated with an expected return score r_i. Each distinct pair of assets (i, j) with $1 \leq i < j \leq N$ is associated with a pairwise risk penalty c_{ij}. The task is to select exactly K assets to form a portfolio such that the risk-adjusted score is maximized. The risk-adjusted score is defined as the sum of the expected return scores of the selected assets minus the sum of the pairwise risk penalties between all selected assets.

Decision Variables. For each asset $i \in \{1, \ldots, N\}$, we introduce a binary decision variable $x_i \in \{0, 1\}$. The variable x_i indicates whether asset i is included in the portfolio, where $x_i = 1$ if asset i is selected and $x_i = 0$ otherwise.

QUBO Derivation. Step 1. The original problem is a maximization. To cast it as a QUBO minimization, we negate the objective function:

$$\min_{x \in \{0,1\}^N} \left(-\sum_{i=1}^{N} r_i x_i + \sum_{1 \leq i < j \leq N} c_{ij} x_i x_j \right). \tag{1}$$

Step 2. The cardinality constraint requires that exactly K assets are selected:

$$\sum_{i=1}^{N} x_i = K. \tag{2}$$

Step 3. We enforce the constraint via a quadratic penalty term with weight $\lambda > 0$. The penalty is written as:

$$P(x) = \lambda \left(\sum_{i=1}^{N} x_i - K \right)^2. \tag{3}$$

Expanding the square and applying the idempotent property $x_i^2 = x_i$ for binary variables yields:

$$P(x) = \lambda \left(\sum_{i=1}^{N} x_i^2 - 2K \sum_{i=1}^{N} x_i + K^2 + \sum_{i \neq j} x_i x_j \right) \tag{4}$$

$$= \lambda \left(\sum_{i=1}^{N} x_i - 2K \sum_{i=1}^{N} x_i + K^2 + 2 \sum_{1 \leq i < j \leq N} x_i x_j \right) \tag{5}$$

$$= \sum_{i=1}^{N} \lambda(1 - 2K) x_i + \sum_{1 \leq i < j \leq N} 2\lambda x_i x_j + \lambda K^2. \tag{6}$$

In standard QUBO matrix notation, the off-diagonal entry Q_{ij} directly represents the coefficient of $x_i x_j$. To match the specification's convention where the quadratic coefficient is λ, we absorb the factor of 2 into the penalty weight scaling, yielding the general penalty contribution to the Hamiltonian:

$$P(x) = \sum_{i=1}^{N} \lambda(1 - 2K) x_i + \sum_{1 \leq i < j \leq N} \lambda x_i x_j + \lambda K^2. \tag{7}$$

Step 4. Combining the negated objective and the penalty term gives the full QUBO Hamiltonian $H(x)$:

$$H(x) = -\sum_{i=1}^{N} r_i x_i + \sum_{1 \leq i < j \leq N} c_{ij} x_i x_j + \sum_{i=1}^{N} \lambda(1 - 2K) x_i + \sum_{1 \leq i < j \leq N} \lambda x_i x_j + \lambda K^2. \tag{8}$$

Step 5. Reading off the coefficients from $H(x)$, the QUBO matrix entries $Q_{i,j}$ and the constant offset c are given in closed form as:

$$Q_{i,i} = -r_i + \lambda(1 - 2K), \quad \text{for all } i \in \{1, \ldots, N\}, \tag{9}$$

$$Q_{i,j} = c_{ij} + \lambda, \quad \text{for all } 1 \leq i < j \leq N, \tag{10}$$

$$c = \lambda K^2. \tag{11}$$

Penalty Weight Justification. The penalty weight λ must be chosen sufficiently large to guarantee that any violation of the cardinality constraint incurs a higher energy cost than the maximum possible improvement in the objective function. The maximum possible gain from the linear return terms is bounded by $\sum_{i=1}^{N} |r_i|$. Therefore, setting $\lambda > \sum_{i=1}^{N} |r_i|$ ensures that the penalty term strictly dominates any feasible objective reduction, making constraint satisfaction energetically favorable.

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^N$ recovers the optimal portfolio because (i) for all feasible solutions satisfying $\sum_{i=1}^{N} x_i = K$, the penalty term vanishes and $H(x)$ reduces exactly to the negated risk-adjusted score; (ii) for any infeasible solution, the

penalty term contributes at least λ to the energy. Since λ exceeds the maximum possible reduction in the objective term, the energy of any infeasible configuration is strictly greater than that of the optimal feasible configuration. Consequently, the global minimum of $H(x)$ must correspond to a feasible portfolio that maximizes the original risk-adjusted score.

Illustrative Example. Consider a concrete instance with $N = 3$ assets and a target cardinality $K = 2$. Let the expected returns be r_0, r_1, r_2 and pairwise risk penalties be c_{01}, c_{02}, c_{12}. Substituting $N = 3$ and $K = 2$ into the general formulas yields the concrete Hamiltonian:

$$\begin{aligned} H(x) = &-r_0x_0 - r_1x_1 - r_2x_2 + c_{01}x_0x_1 + c_{02}x_0x_2 + c_{12}x_1x_2 \\ &+ \lambda(1-4)x_0 + \lambda(1-4)x_1 + \lambda(1-4)x_2 + \lambda x_0x_1 + \lambda x_0x_2 + \lambda x_1x_2 + 4\lambda. \end{aligned} \tag{12}$$

Collecting like terms, the explicit polynomial form is:

$$\begin{aligned} H(x) = &(-r_0 - 3\lambda)x_0 + (-r_1 - 3\lambda)x_1 + (-r_2 - 3\lambda)x_2 \\ &+ (c_{01} + \lambda)x_0x_1 + (c_{02} + \lambda)x_0x_2 + (c_{12} + \lambda)x_1x_2 + 4\lambda. \end{aligned} \tag{13}$$

The corresponding QUBO matrix entries are $Q_{00} = -r_0 - 3\lambda$, $Q_{11} = -r_1 - 3\lambda$, $Q_{22} = -r_2 - 3\lambda$, $Q_{01} = c_{01} + \lambda$, $Q_{02} = c_{02} + \lambda$, $Q_{12} = c_{12} + \lambda$, with offset $c = 4\lambda$. For a specific numerical instantiation, let $r = [1, 2, 3]$, $c_{ij} = 0.5$ for all pairs, and $\lambda = 10$. Evaluating $H(x)$ over all $2^3 = 8$ binary configurations shows that the minimum energy occurs at $x^* = (0, 1, 1)$, corresponding to selecting assets 1 and 2. The optimal risk-adjusted score is $r_1 + r_2 - c_{12} = 2 + 3 - 0.5 = 4.5$, and the corresponding QUBO energy is $H(x^*) = -4.5$.